SOFIANE MALLEM

Backend Developer · Python

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LinkedIn · GitHub · Portfolio

SUMMARY

Python developer with a year of production experience running data and rendering pipelines on a Linux server fleet. Builds backend services with FastAPI, PostgreSQL and Redis and deploys them with Docker, Kubernetes and CI/CD.

EXPERIENCE

Python Developer, *Synthavo – Stuttgart* **Nov 2025 – Apr 2026**

- Owned the CAD data pipeline (Python, Pandas): validated, transformed and uploaded customer datasets feeding a Blender rendering service on 5+ Linux servers.
- Migrated rendering to a newer Blender: 41% faster end-to-end rendering on top of prior optimizations, 99%+ uptime via monitoring and bottleneck fixes.
- Maintained the rendering-to-training pipeline and ran YOLOv5 training end to end; delivered weights to QA.
- Ran deployments with Ansible (existing IaC); served as primary contact for production incidents and customer requirements.

Python Developer Intern, *Synthavo – Stuttgart* **Feb 2025 – Aug 2025**

- Built Python tooling for CAD validation and transformation, improving rendering reliability.
- Added STL mesh reduction ahead of rendering, freeing RAM for more parallel instances and making end-to-end rendering 37% faster.
- Cut customer onboarding provisioning from ∼5 days to 2–3 through Python automation.

PROJECTS

Hang | *Real-Time Messaging Platform, Horizontally Scaled* **GitHub · 2026**
Stack: FastAPI · WebSockets · OAuth · Redis Pub/Sub · PostgreSQL · Kubernetes (kind, Kustomize) · Prometheus/Grafana · React/TS

- Created a dating/chat platform: Google OAuth, S3 uploads, swiping, typing indicators, read receipts and cursor pagination.
- Synced WebSocket events across stateless FastAPI replicas via Redis Pub/Sub; deployed to dev/qa/prod Kustomize overlays on kind with probes, a dedicated prod node pool and Prometheus/Grafana monitoring.
- CI deploys to kind and sends a WebSocket message through the ingress as a smoke test; version tags publish images to GHCR.
- Load-tested 190,000 messages through the ingress with no loss at 2,000 concurrent WebSockets; found and fixed a DB-session leak and a join-before-accept race.

LAD (Local AI Daemon) | *Self-Hosted AI Assistant with Tool-Calling Agent* **GitHub · 2026**
Stack: FastAPI · React/TS · PostgreSQL + pgvector · Ollama · MCP · SSE · Docker Compose · Nginx · GitHub Actions

- Developed a local AI assistant on Ollama (no external API): streaming chat, file uploads, MCP tool-calling agent with server-side per-mode tool allowlists.
- Sandboxed shell execution in an offline, non-root container with dropped capabilities and resource limits.
- Implemented cross-chat memory from embedded rolling summaries (pgvector); tested with pytest (real Postgres) and Vitest/MSW, with 3 CI workflows.

Durden | *Object Detection API on Google Cloud Run* **GitHub · Live · 2026**
Stack: FastAPI · Ultralytics YOLO · Docker · GitHub Actions · GHCR · Google Cloud Run · Workload Identity Federation

- Built a YOLO inference API with per-request model selection, background model loading and readiness endpoints.
- Set up CI that runs lint and tests, publishes to GHCR and deploys to Cloud Run via workload identity federation (no stored keys).

EDUCATION

42 Heilbronn (peer-learning software engineering program) **Feb 2023 – Nov 2025**
Software Engineering Course

University of Science and Technology Houari Boumediene (USTHB) **Sep 2017 – Jun 2020**
Bachelor of Computer Science

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript, React, C, C++, Bash
Backend: FastAPI, REST APIs, WebSockets, OAuth, Async Python, Redis Pub/Sub
Data: PostgreSQL, Redis, Pandas, Data Pipelines
Infrastructure: Docker, Kubernetes, Linux, Nginx, CI/CD, GitHub Actions, Google Cloud Run, Ansible, Prometheus, Grafana
Practices: Git, pytest, Load Testing, Unit/Integration Testing

LANGUAGES

English (Fluent) | French (Native) | German (A1, learning)